

TypeError: Invalid input argument for exponent. (tfq.convert_to_tensor)

<https://github.com/tensorflow/quantum/issues/712>

ROW 131

TypeError: Invalid input argument for exponent. (tfq.convert_to_tensor) #712

[New issue](#)[Closed](#)

Shuhul24 opened on Aug 19, 2022 · edited by Shuhul24

Edits · ...

I have written following code:

```
def convert_to_circuit(image):
    values = tf.reshape(image, [-1])
    qubits = cirq.GridQubit.rect(4, 4)
    circuit = cirq.Circuit()
    for i in range(4):
        for j in range(4):
            k = (4 * i) + j
            circuit.append(cirq.H(cirq.GridQubit(i, j)))
            circuit.append(cirq.Rx(rads=values[k]).on(cirq.GridQubit(i, j)))
    return circuit
```

Here, the input in the function is a tensor. Basically, I have considered the image dataset into a tensor instead of a numpy.ndarray.

After this function, I wrote the following block of code and raised an error:

```
input_dis = [convert_to_circuit(x) for x in img]
input_disc = tfq.convert_to_tensor(input_dis)
```

The error is:

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-58-49439858826a>:1(https://localhost:8080/#) in <module>
      3 # print(type(input_dis))
      4 # print(type(input_dis[0]))
----> 5 input_disc = tfq.convert_to_tensor(input_dis)

10 frames
[/usr/local/lib/python3.7/dist-packages/tensorflow_quantum/core/serialize/serializer.py](https://localhost:8080/#)
    88     """This is the second extractor for above."""
    89     if not isinstance(x, (numbers.Real, sympy.Expr)):
----> 90         raise TypeError("Invalid input argument for exponent.")
    91
    92     if isinstance(x, numbers.Real):

TypeError: Invalid input argument for exponent.
```

Can you please help me out why is this happening? The only change I did was taking input the tensor instead of numpy.ndarray.



Shuhul24 changed the title **TypeError: Invalid input argument for exponent. (tfq.convert_to_tensor)** to **TypeError: Invalid input argument for exponent. (tfq.convert_to_tensor)** on Aug 19, 2022



lockwo on Aug 19, 2022

Contributor · ...

You can't use a tensor to specify rotations, it must be a numbers.Real or sympy.Expr. Simply cast the tensor to a numpy array to solve this problem. This worked for me (I actually don't know whether casting to numpy as a single value or as an array is more efficient since I'm not sure what that looks like in the underlying memory operations, but may be worth considering if you iterate a lot over this function), where the key difference is `rads=values[k].numpy()`:

```
import tensorflow_quantum as tfq
import cirq
import tensorflow as tf

def convert_to_circuit(image):
    values = tf.reshape(image, [-1])
    qubits = cirq.GridQubit.rect(4, 4)
    circuit = cirq.Circuit()
```

Can you please help me out why is this happening? The only change I did was taking input the tensor instead of numpy.ndarray.



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import cirq
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def convert_to_circuit(image):
    values = tf.reshape(image, [-1])
    qubits = cirq.GridQubit.rect(4, 4)
    circuit = cirq.Circuit()
    for i in range(4):
        for j in range(4):
            k = (4 * i) + j
            circuit.append(cirq.H(cirq.GridQubit(i, j)))
            circuit.append(cirq.Rx(rads=values[k].numpy()).on(cirq.GridQubit(i, j)))
    return circuit

img = tf.random.uniform(shape=[1, 16])
input_dis = [convert_to_circuit(x) for x in img]
input_disc = tfq.convert_to_tensor(input_dis)
```



lockwo on Aug 24, 2022

Contributor ...

I believe [#713](#) resolved this.



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Shuhul24 on Aug 24, 2022

Author ...

Thanks for this as well !



Shuhul24 closed this as **completed** on Aug 24, 2022

lockwo mentioned this on Aug 24, 2022

[About the input of quantum neural network in tensorflow quantum #377](#)